

Ten-Layer Superloop–Anchor Frontier Review

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Abstract

RH-242–RH-251 connect cloud-extracted traces to noisy periodic loops, define the deterministic numerator anchor, and identify the exact frontier of the current cloud class. Fixed-noise superloop and orthogonal quotient identities are exact, and a block-power theorem gives the correct conditional route to an all-order envelope. Conversely, the frozen candidate window fails the anchor test even after convex relaxation, while unbounded weights produce non-spectral multiplicity explosion. No complete finite-head/analytic-tail certificate exists. We classify the ten layers, preserve the genuine local progress, and keep Gates A–E open.

1 Batch decision

The route coordinate after RH-251 is

exact_superloop_quotient_frozen_anchor_class_obstructed
open_new_selector_uniform_tail.

This wording has two deliberate halves. “Exact superloop quotient” records real structural progress. “Frozen anchor class obstructed” prevents the finite negative result from being inflated into global cloud nonexistence.

The batch contains 16,191 explicitly audited loop, trace, coefficient, prefix, endpoint, and order items, with zero review-audit failures. A single LP dual additionally certifies a binary class containing 139,572,890 shell subsets. These are finite facts; none is an all-order or continuum theorem.

2 The ten layers

paper	layer	result	first missing theorem
RH-242	periodic-loop realization	Exact graded physical-loop/atomic-counterloop identity; 14,584 enumerated loops and 352 signed residuals	uniform loop grouping and coefficient anchor
RH-243	deterministic numerator anchor	Exact one-step Hardy-scaled coefficient dictionary and even two-step projection	identification of those coefficients in a selected noisy cloud
RH-244	anchored prefixes	Equal-radius target balls are disjoint; 0/32 passes among 543 prefixes	a non-prefix anchored selector
RH-245	orthogonal quotient	Exact $\tau_n = \text{Tr } C^n$ with a norm-one orthogonal projection	uniform stability of the moving algebraic root space
RH-246	block-power envelope	Exact m -block criterion; finite $q_{12} = 0.393300$ on 17 endpoints	noise-uniform K_m, η_m, L_r and finite head
RH-247	absolute majorant	Perron sector forces root rate at least $r_H^{-1} > 1$	cancellation-preserving signed quotient estimate

paper	layer	result	first missing theorem
RH-248	shell zonotope	Full single-use shell box has 0/32 passes with dual gap $< 8 \times 10^{-15}$	expanded window or structurally signed selector
RH-249	nonnegative cone	6/32 remain unreachable; 26 fits need maximum weights $40.58\text{--}5.80 \times 10^{10}$	legal algebraic multiplicity and operator realization
RH-250	head/tail gluing	Exact determinant error interface; zero complete present certificates	anchored head, uniform quotient tail, and target tail
RH-251	frontier review	local fixed-noise progress retained; frozen reweighting route stopped	a genuinely new anchor/selector or uniform theorem

3 Exact progress that survives

Periodic superloops. RH-242 proves for every fixed noisy operator and selected algebraic multiset that

$$\tau_n = \text{Tr } A_\sigma^n - \sum_{s \in S_\sigma} s^n = \text{Str}((A_\sigma \oplus S_\sigma)^n), \quad n \geq 2,$$

and realizes the physical term as a cyclic Gaussian loop integral [1]. The identity is exact; the continuum selector is still an assumption.

Independent coefficient anchor. RH-243 separates the all-order size problem from the numerator identity. If P_n is the deterministic flat trace, then

$$a_n = r_H^{-n} \left[P_n - 1 - (-1)^n + 2\mathbf{1}_{\{2|n\}} \lambda^{-n/2} \right],$$

and the symmetric two-step coefficient is $b_k = a_{2k}$ [2]. This defines the target but does not put it in the cloud.

Cancellation-preserving quotient. RH-245 replaces the ill-conditioned Riesz estimate by an orthogonal quotient. For the invariant generalized root space E , $Q = I - \Pi_E$, and $C = QAQ|_{E^\perp}$,

$$\tau_n = \text{Tr } C^n.$$

The projection has norm one; $E^\perp$ need not reduce A [4]. RH-246 then proves that one uniformly contractive power C^m gives the geometric trace envelope and an explicit logarithmic tail [5]. These theorems survive even though their uniform hypotheses are not yet certified.

Determinant interface. RH-250 proves that, on $|z| \leq R$, the logarithmic determinant error is bounded by the anchored finite-head error plus the quotient tail plus the target tail [9]. This is the exact interface to a future RH-240 application; the present three certificates are incomplete.

4 Obstructions and their scope

The zero-target and anchored problems cannot be conflated. The anchor norm through order 12 is 0.4945054357, whereas all archived tolerances are at most 0.04; the two equal-radius balls are disjoint. Exhausting the frozen prefix class gives 0/32 passes [3].

RH-248 removes the radial-order excuse. It places every complete shell in a box $0 \leq w_j \leq 1$ and dualizes the anchored weighted ℓ^1 distance. The box itself has distance 0.146498–0.424018, at least 10.17 times the local tolerance, so every prefix, interval, and binary subset fails [7]. This is a frozen-window statement.

Allowing $w_j \geq 0$ without an upper bound does not produce a credible rescue. Six endpoints remain outside the cone; the other 26 need minimum maximum weights between 40.584 and 5.8018×10^{10} [8]. These are arbitrary real repetitions, not algebraic spectral multiplicities.

The long-loop estimate also has a strict firewall. Any majorant that takes absolute values of physical and atomic sectors before cancellation contains $|p_\sigma|^n = r_H^{-n}$, so its root rate is superunit. All 352 archived order cases confirm rates 1.2532 – 3.5568 [6]. A viable proof must work after signed/complex grouping.

5 Gate audit

gate	requirement	status after RH-251
A	canonical intrinsic dynamical determinant and identification	open
B	time-oriented scattering or unitary completion	untouched/open
C	self-adjoint generator and intrinsic $T \log T$ law	untouched/open
D	von Mangoldt-weighted prime-power traces	untouched/open
E	equality with the completed-zeta spectral divisor	untouched/open

The exact identities are local Gate-A infrastructure, not Gate-A closure. There is no locally uniform anchored relative determinant family, no Hilbert–Polya operator, no Riemann-zero identification, no zeta-divisor equality, and no implication for RH.

6 Next-batch reopening triggers

The same frozen shell window should not be reweighted again. A new batch is justified by at least one genuinely new input:

1. a larger resolved spectral window, followed immediately by an anchored zonotope/cone audit rather than a zero-target fit;
2. a signed or complex selector derived from invariant quotient structure, not arbitrary real shell weights;
3. a noise-uniform theorem for one quotient block power, including the finest discretizations and a continuum bridge; or
4. an analytic all-order tail estimate for the deterministic numerator target.

Until then, the honest endpoint is a scoped route stop with exact fixed-noise progress preserved.

References

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